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Aim: Demo of OpenAI/TensorFlow Tools

 Explore and experiment with OpenAI or TensorFlow tools and libraries.

 Perform a demonstration or mini-project showcasing the capabilities of the tools.

 Discuss and present the findings and potential applications.

Code

```
from flask import Flask, render_template, request, jsonify
import json
import urllib.request
import urllib.error

app = Flask(__name__)

OLLAMA_URL = "http://localhost:11434/api/chat"
MODEL = "llama3.2:1b"

SYSTEM_PROMPT = (
    "You are Nova AI, a friendly educational assistant. "
    "Answer clearly, accurately, and in simple language."
)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/chat", methods=["POST"])
def chat():
    data = request.get_json(silent=True) or {}
    messages = data.get("messages", [])
```

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```
    if not isinstance(messages, list) or not messages:
        return jsonify({"error": "Please enter a message."}), 400

    safe_messages = []
    for message in messages[-20:]:
        if not isinstance(message, dict):
            continue
        role = message.get("role")
        content = message.get("content")

        if role not in ("user", "assistant"):
            continue
        if not isinstance(content, str):
            continue

        content = content.strip()
        if content:
            safe_messages.append({
                "role": role,
                "content": content[:5000]
            })

    if not safe_messages or safe_messages[-1]["role"] != "user":
        return jsonify({"error": "Please send a valid message."}), 400

    payload = {
        "model": MODEL,
        "messages": [
            {"role": "system", "content": SYSTEM_PROMPT},
            *safe_messages
        ],
```

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```
    "stream": False
}

req = urllib.request.Request(
    OLLAMA_URL,
    data=json.dumps(payload).encode("utf-8"),
    headers={"Content-Type": "application/json"}
)

try:
    with urllib.request.urlopen(req, timeout=180) as response:
        result = json.loads(response.read().decode("utf-8"))

    return jsonify({
        "reply": result["message"]["content"]
    })

except urllib.error.HTTPError:
    return jsonify({
        "error": "Ollama returned an error. Check that the model is installed."
    }), 502

except (urllib.error.URLError, TimeoutError):
    return jsonify({
        "error": "Cannot connect to Ollama. Make sure it is running."
    }), 503

except (KeyError, json.JSONDecodeError):
    return jsonify({
        "error": "Could not read the AI response."
    }), 502
```

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```
if __name__ == "__main__":
    app.run(debug=False, host="127.0.0.1", port=5000)
```

chatbot code

```
import json
import urllib.request
import urllib.error

URL = "http://localhost:11434/api/chat"
MODEL = "llama3.2:1b"

messages = [
  {
    "role": "system",
    "content": (
      "You are a friendly AI chatbot for college students. "
      "Explain concepts clearly using simple language."
    )
  }
]

print("=" * 45)
print("        MY LOCAL AI CHATBOT")
print("=" * 45)
print("Powered by Ollama and Llama 3.2")
print("Type 'exit' to close the chatbot.")

while True:
    user_input = input("\nYou: ").strip()
```

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```
if user_input.lower() == "exit":
    print("Bot: Goodbye! Have a great day!")
    break

if not user_input:
    print("Bot: Please enter a message.")
    continue

messages.append({
    "role": "user",
    "content": user_input
})

data = json.dumps({
    "model": MODEL,
    "messages": messages,
    "stream": False
}).encode("utf-8")

request = urllib.request.Request(
    URL,
    data=data,
    headers={"Content-Type": "application/json"}
)

try:
    with urllib.request.urlopen(request, timeout=180) as response:
        result = json.loads(response.read().decode("utf-8"))

    answer = result["message"]["content"]
```

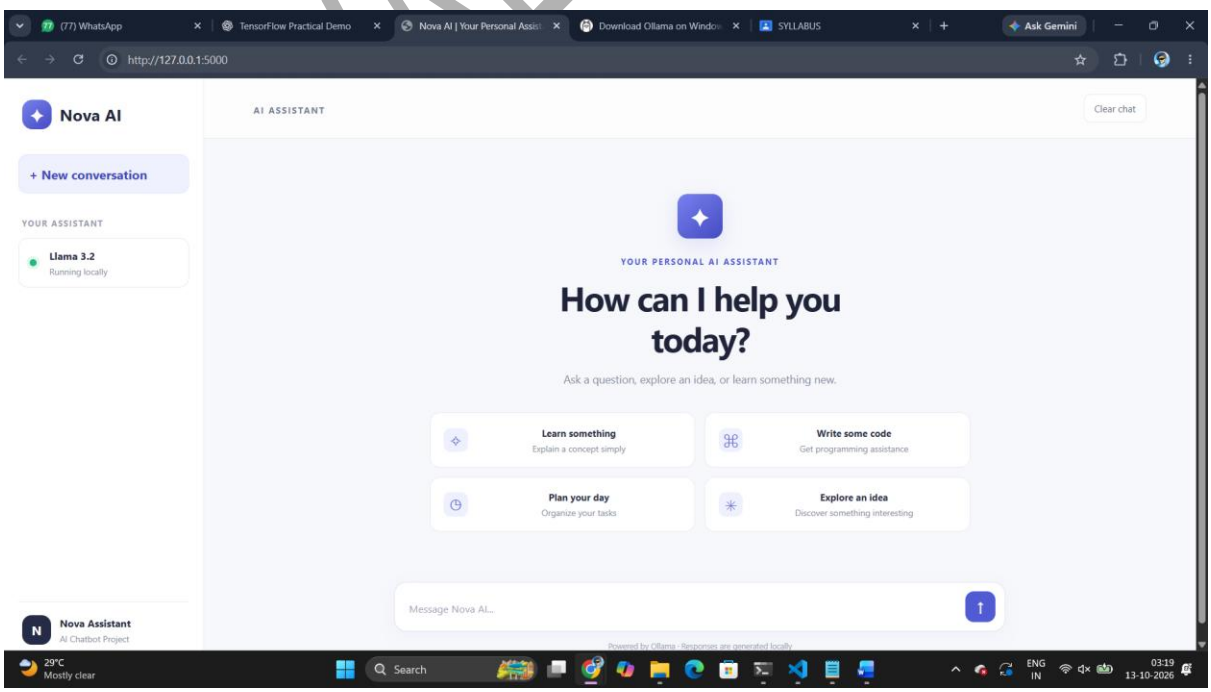
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```
    print("\nBot:", answer)

    messages.append({
        "role": "assistant",
        "content": answer
    })

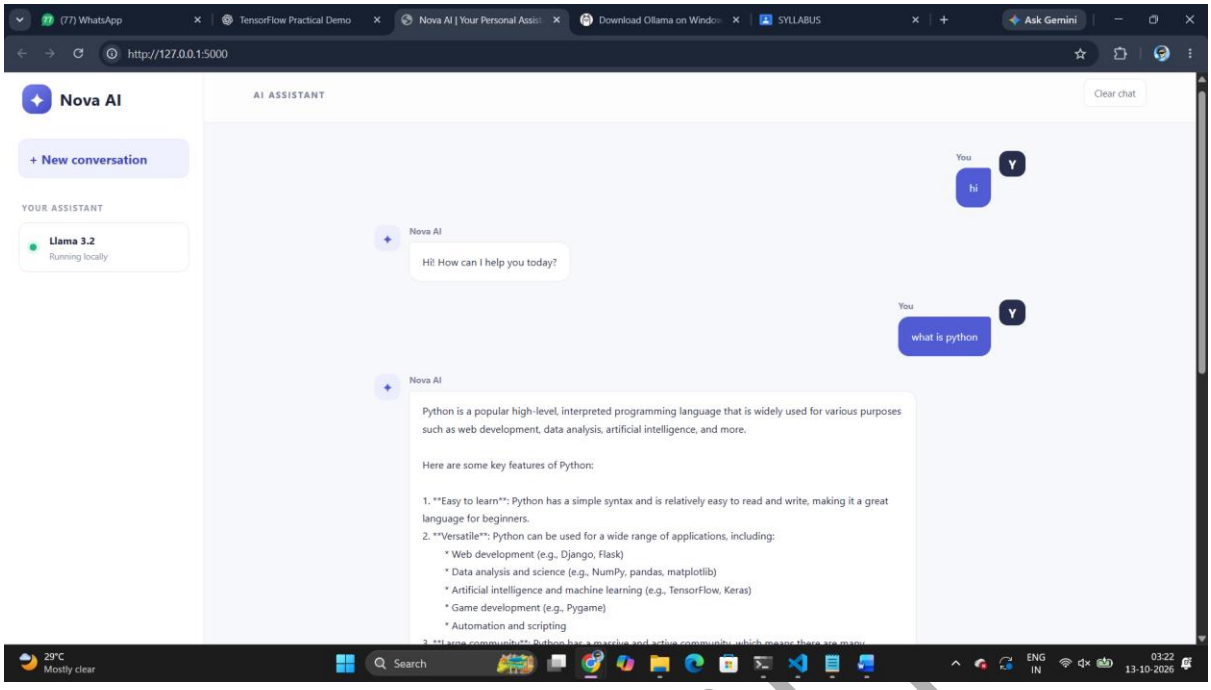
except (urllib.error.URLError, TimeoutError) as error:
    messages.pop()
    print("\nCould not connect to Ollama.")
    print("Check that Ollama is running.")
    print("Details:", error)

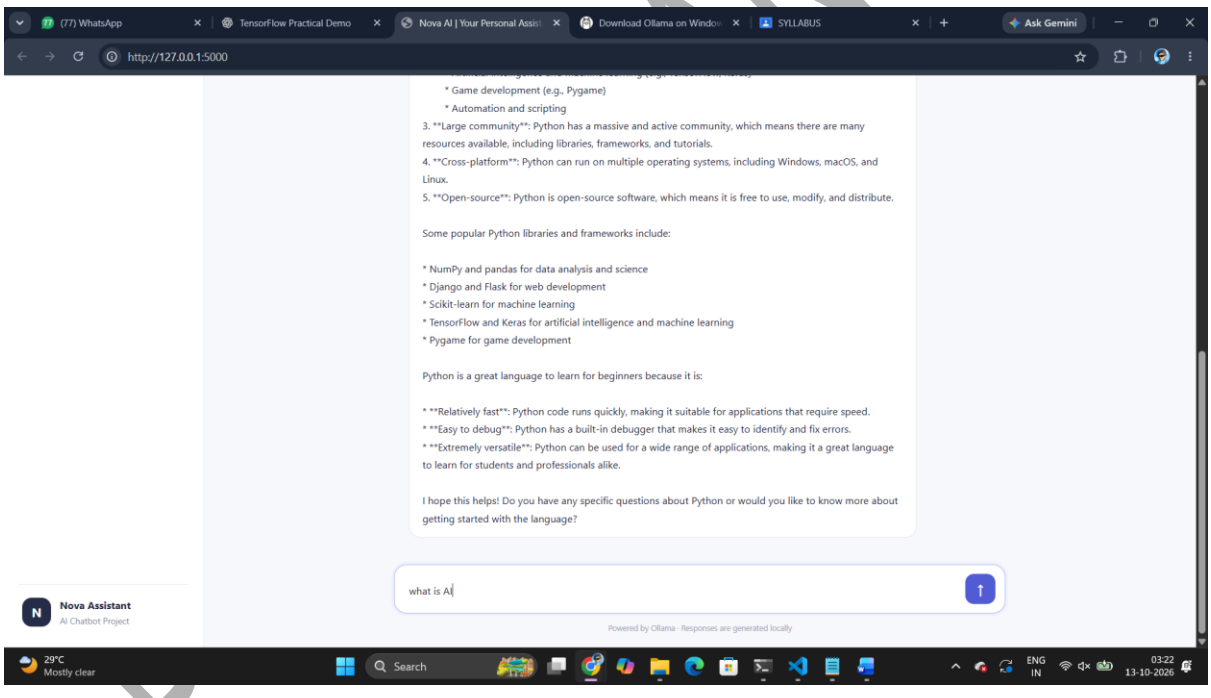
except (KeyError, json.JSONDecodeError) as error:
    messages.pop()
    print("\nCould not read the model response.")
    print("Details:", error)
```



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